

Similarity Learning for Image Representations

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Abstract

We propose novel methods for knowledge distillation and representation learning for computer vision by explicitly learning a similarity function between images. Our approach stands in contrast to standard knowledge distillation, where a student neural network learns to emulate the teacher network’s output scores, and to recent self-supervised contrastive representation learning methods, which learn to score augmented versions of the same image higher than a pair of distinct images. Our preliminary results for similarity-based distillation on CIFAR-10 are encouraging, as our ResNet-18 student surpasses its ResNet-50 teacher. Distillation results on ImageNet are pending and similarity-based representation learning remains in development.

The goal of this work is to train a deep neural network f that embeds images into $\mathbb{R}^d$. The resulting embeddings are then applied to supervised linear classification, where the objective is to learn a matrix $W \in \mathbb{R}^{C \times d}$ so that the index of the largest entry in $Wf(x)$ is equal to the label $y \in \{1, \dots, C\}$ (e.g. “car”, “dog”) for an input image x .

Embeddings are said to generalize well if they can successfully be applied to a multitude of classification tasks, and recent work in contrastive self-supervised contrastive representation learning (Oord et al., 2018; He et al., 2020; Chen et al., 2020) introduced algorithms for learning image representations of this calibre. The key is learning to score the similarity between two images. Two views of the same image x, x_1 (a “positive sample”) should be assigned a higher similarity score than two distinct images x, x_k , $k \in \{2, \dots, m\}$ (a “negative sample”), as reflected by the objective

$$\operatorname{argmin}_f - \log \left[\frac{\exp(\mu(f(x), f(x_1)))}{\sum_{i=1}^m \exp(\mu(f(x), f(x_i)))} \right], \quad (1)$$

where μ is a function (e.g. cosine similarity) giving the model’s notion of similarity. The batch size m is usually large (e.g. 8192), giving a large number of negative samples per positive sample.

We remark that the above objective is based solely on similarity scores assigned to image pairs and is independent of any classification task. Thus, the success of the above works shows that similarity alone is sufficient to learn representations that generalize.

Our contribution is to propose a novel approach for learning image representations from similarity. We explicitly specify the similarity score that our learner should output for a given pair of images, rather than teaching it to learn any function that ranks certain pairs above others. Our approach is simple and requires no negative samples, allowing for faster training. We use insights from word embeddings as our inspiration. The popular word2vec algorithm (Mikolov et al., 2013) had a contrastive loss in its original formulation, and was shown to be equivalent to the low-rank factorization of a matrix of dataset statistics (Levy and Goldberg, 2014). In the same way, learning embeddings with our method corresponds to explicitly factorizing a matrix of similarity values into a collection of vector representations of images.

Using these ideas, we also explore whether learning from similarity alone is sufficient for knowledge distillation, where the goal is to train a “student” network to mimic the predictions of the (usually larger) pre-trained “teacher” network on a specific classification task (Hinton et al., 2015). We propose to instead have the student learn representations to mimic the teacher’s notion of similarity between images.

Let f denote the student model and g denote the teacher model. Let $\mathcal{B}$ be the collection of batches in the training set. Let $x_1^{(b)}, \dots, x_m^{(b)}$ be the images in a batch b . Then, our similarity-based distillation loss is

$$L_d(f) = \sum_{b \in \mathcal{B}} \sum_{i,j=1}^m \left(\cos(f(x_i^{(b)}), f(x_j^{(b)})) - \cos(g(x_i^{(b)}), g(x_j^{(b)})) \right)^2. \quad (2)$$

Unlike the original knowledge distillation formulation, this makes no assumption about a downstream classification task, thus allowing for the potential distillation of generalizable features.

For similarity-based representation learning from scratch, we explicitly define the similarity function $s(x, x')$ to be learned. It is a decreasing function of the strength α required to map an image x to another image x' via some transformation T_α (it is independent of x, x'). For training, we compose the same transformations as in Chen et al. (2020): Gaussian blur, colour jitter, and random crop. α controls the standard deviation of the blur, the intensity of the jitter, and the proportion of the image which is cropped out. Given a distribution τ , our proposed loss is

$$L_s(f) = \mathbb{E}_{\alpha \sim \tau} \left(\cos(f(x_i), f(T_\alpha(x_i))) - s(x_i, T_\alpha(x_i)) \right)^2. \quad (3)$$

Note that this loss does not require negative samples, which we can view as being implicitly captured by the case where α is large.

Preliminary results for distillation on the CIFAR-10 dataset (Krizhevsky et al., 2009) are extremely encouraging. We use the popular ResNet neural network architecture (He et al., 2016). The teacher is a ResNet-50, and we evaluate both with a student of the same size, and with a smaller ResNet-18 student. Our approach outperforms the standard distillation method of Hinton et al. (2015). Results for ImageNet (Russakovsky et al., 2015) are pending.

Model	Method	Train Accuracy	Validation Accuracy
Teacher Baseline	-	99.99	94.64
ResNet-50	KD	98.81	94.12
ResNet-50	Ours	98.82	95.57
ResNet-18	Ours	98.64	95.55

Table 1: Summary of distillation results on the CIFAR-10 dataset. KD designates the standard distillation approach (Hinton et al., 2015).

Similarity-based representation learning remains in development as we continue work on selecting a suitable distribution τ for the augmentation strength and the function s to be learned. As a starting point, we have compiled the similarity scores output by the SimCLR model (Chen et al., 2020) for various choices of α .

We conjecture that similarity is a fundamental property of learning, and that with a proper choice of s , we can learn representations that generalize from similarity alone, without the need for class labels or negative samples. In future work, we will complete the development of our similarity-based embedding learning algorithm to empirically test this conjecture.

References

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